
INTERNATIONAL GCSE **CHEMISTRY**

9202/1

Paper 1

Mark scheme

November 2019

Version: 1.0 Final Mark Scheme

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqaexams.org.uk

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error/contradiction negates each correct response. So, if the number of errors/contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols/formulae

If a student writes a chemical symbol/formula instead of a required chemical name, full credit can be given if the symbol/formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(...) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

'Ignore' is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

'Do **not** accept' means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
01.1	3		1	AO2 3.1.2b	G
01.2	monomers		1	AO1 3.10.2a	A
01.3	H H H H H H H H —C— C— C— C— C— C— C— C— H Cl H Cl H Cl H Cl	1	AO2 3.10.2a	A	
01.4	any one from: • litter • shortage of landfill space • harmful to wildlife • pollution of oceans • pollution of atmosphere (if incinerated)		1	AO1 3.10.2e	E
01.5	D		1	AO3 3.10.2d	A
01.6	thermosetting		1	AO1 3.10.2.c	E
01.7	crosslinks (between polymer chains)		1	AO1 3.10.2.c	E
Total			7		

Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
02.1	labelled with copper		1	AO2 3.3.1b	E
	bar drawn correctly to 6.8 g	allow a tolerance of $\pm \frac{1}{2}$ a small square	1		
02.2	(total mass of coin) = 9.5 (g) $\frac{6.8}{9.5} (\times 100)$ = 71.58 72%	an answer of 72(%) scores 4 marks an answer of 71.57894737 (%) correctly rounded to more than 2 sig figs scores 3 marks	1	AO2 3.3.1b	E
		allow correct use of incorrectly calculated total mass of coin	1		
			1		
		allow an answer correctly rounded to 2 significant figures from an incorrect attempt of the percentage calculation	1		
02.3	can be bent		1	AO1 3.3.1c	A
	unreactive		1		
02.4	delocalised		1	AO1 3.2.2g	G
02.5	iron sulfate + copper	either order	1	AO2 3.3.1.1.h 3.6.1a	E
02.6	iron is more reactive (than copper)		1	AO2 3.3.1.1.b 3.3.1.1.h	E

02.7	any one from: - scrap iron is cheap - scrap iron is abundant - avoids disposal of scrap iron		1	AO2 3.3.1.1h 3.3.1.1i	E
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02.8			2	AO1 3.3.1.1g 3.3.2b 3.3.2c	G
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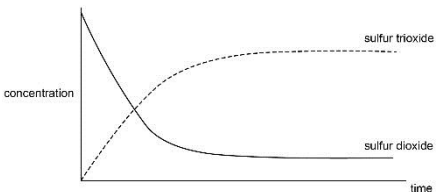
02.9	reduced		1	AO1 3.3.2d	G
	gained		1	3.3.2e	

Total			16		
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Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
03.1	atoms		1	AO1 3.6.1c	G
03.2	(moles Fe = $\frac{2.24}{56}$) = 0.04 (M_r of FeS) = 88 (0.04 × 88) = 3.52 (g) alternative approach: (M_r of FeS) = 88 (1) (mass of FeS =) $2.24 \times 88/56$ (g) (1) = 3.52 (g) (1)	an answer of 3.52 g scores 3 marks allow correct calculation using an incorrectly calculated value of moles Fe and/or M_r of FeS allow correct use of an incorrectly calculated value of M_r of FeS	1 1 1	AO2 3.6.1d	E
03.3	mass (of tube and contents) decreased but the mass should have stayed constant		1 1	AO3 3.6.1c 3.6.1e	E
03.4	one of the substances boiled / evaporated or a different reaction took place (so) mass was lost (as a gas)	allow a gas was released MP2 must be linked to MP1	1 1	AO3 3.6.1c 3.6.1e	E
Total			8		

Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
04.1	any one from: - measuring cylinder - pipette - burette	do not accept beaker	1	AO4 3.9.2c	E
04.2	change any one from: - lag / insulate beaker - change for plastic beaker - add a lid		1	AO4 3.9.2c	E
	reason reduce energy loss	allow reduce heat loss	1		
04.3	stir (the mixture)	allow to distribute energy evenly (through the mixture)	1	AO4 3.9.2c	E
	ensure temperature is even (through the mixture) or to complete the reaction quickly		1		
04.4	0.1 (°C)		1	AO3 3.9.2c	G
04.5	(25.9 – 20.1) = 5.8 (50 × 4.2 × 5.8) = 1218 (J) 1.218 (kJ)	an answer of 1.218 (kJ) or 1220 (kJ) or 1.2 (kJ) scores 3 marks an answer of 1218 (J) 1200 (J) or 1220 (J) scores 2 marks	1	AO2 3.9.2c	E
		allow correct use of an incorrectly calculated temperature change	1		
		do not accept 1000	1		

04.6	$\left(\text{moles} = \frac{4.14}{138}\right) = 0.03$ $\left(\text{energy released per mol}\right) = \frac{1.035}{0.03}$ or 34.5 $\Delta H = -34.5 \text{ (kJ/mol)}$	an answer of -34.5 or -35 (kJ/mol) scores 3 marks must have negative sign for this mark	1 1 1	AO2 3.6.3a 3.9.2 b 3.9.1a	E
04.7	50 (kJ)	do not accept -50	1	AO2 3.9.2d	E
04.8	130 (kJ)		1	AO2 3.9.2d	E
Total			14		

Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
05.1	pressure	ignore catalysts	1	AO1 3.8.2b	E
	temperature		1	3.8.2c 3.8.2d	
05.2	gradient / slope decreases		1	AO3 3.8.1a	E
05.3	number of (reactant) particles in the same volume decreases	allow reactant particles are further apart	1	AO1 3.8.1d	E
	(so) frequency of collisions decreases	ignore fewer particles ignore fewer collisions	1		
05.4	a line that starts at origin and mirrors sulfur dioxide line		1	AO3 3.8.1a	E
05.5	(rates of forward and backwards reactions are) the same		1	AO1 3.8.2a	E
05.6		an answer of 16.7 or 16.664 or 16.66 (mol/dm ³) scores 4 marks answer of 16700 or 16664 (mol/dm ³) scores 3 marks		AO2 3.6.4d 3.6.4a	E
	(vol =) 5×10^4 or 50 000 dm ³		1		
	$(\text{mol} = \frac{50000}{24}) = 2083 \text{ (mol)}$	allow correct use of incorrect / no conversion of volume	1		
	$(\text{conc} = \frac{\text{mol}}{\text{vol}}) = \frac{2083}{125}$	allow correct use of incorrect calculation of moles	1		
	16.7 (mol/dm ³)	allow 16.664 (mol/dm ³) correctly rounded to at least 2 sig figs	1		

		alternative approach for mark points 1 and 2 $24 \text{ dm}^3 = 24\,000 \text{ cm}^3$ (1) (amount of $\text{SO}_3 = 5 \times 10^7 / 24\,000 =$) 2083 (mol) (1)			
Total			11		

Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
06.1	3 pairs of electrons in overlap 1 non-bonding pair on each N		1 1	AO1 3.2.1g	E
06.2	$\text{N}_2 + 2 \text{O}_2 \rightarrow 2 \text{NO}_2$		1	AO2 3.6.1a	E
06.3	acid rain	allow smog formation allow effects of acid rain allow respiratory problems	1	AO1 3.10.1.2d	E
06.4	insufficient oxygen (so) incomplete combustion occurs	allow insufficient air	1 1	AO1 3.10.1.2d	E
06.5	carbon dioxide	allow CO_2	1	AO2 3.10.1.2d	G
06.6	1 – 100 nm	allow a few hundred atoms	1	AO1 3.2.4	E
06.7	any one from: • larger surface area so less catalyst needed • thinner layer of catalyst	allow same amount provides a larger surface area	1	AO2 3.2.4	E
Total			9		

Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
07.1	they have the same number of electrons in their highest energy level	allow they are in the same group ignore they are non-metals	1	AO1 3.1.3b	E
07.2	covalent		1	AO1 3.2.1h	E
07.3	silicon dioxide is a giant structure (so) all the atoms in (silicon dioxide has) are linked by strong (covalent) bonds carbon dioxide is a simple molecule (carbon dioxide has) weak intermolecular forces (silicon dioxide is a solid so) there is insufficient energy (at room temperature) to break the bonds or (carbon dioxide is a gas because) there is enough energy (at room temperature) to overcome intermolecular forces	allow (silicon dioxide has) many strong (covalent) bonds allow weak intermolecular bonds allow more energy needed to break bonds in SiO ₂ than overcome intermolecular forces in CO ₂	1 1 1 1 1	AO1 3.2.2d 3.2.2f	E
Total			7		

Question	Answers	Mark	AO/ Spec. Ref	ID
08	Level 3: A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.	5–6	AO3 AO1 3.7.2a 3.7.2b	E
	Level 2: Some logically linked reasons are given. There may also be a simple judgement.	3–4		
	Level 1: Relevant points are made. They are not logically linked.	1–2		
	No relevant content	0		
	Indicative content • G and L are transition metals because: ◦ transition metals have ions with different charges ◦ G forms chlorides containing the ions G^+ and G^{2+} ◦ L forms chlorides containing the ions L^{2+} and L^{3+} • transition metals form coloured compounds ◦ G forms a blue-green chloride ◦ L for green and yellow chlorides • transition metals have higher melting points ◦ 1083 and 1535 °C • transition metals have higher densities ◦ 8.92 and 7.86 g/cm³ • transition metals are stronger and harder ◦ 2.5 and 4.5 arbitrary units • transition metals are less reactive (than Group 1 metals) and do not react as vigorously with water or oxygen ◦ no reaction or very slow reaction with water • J could be a Group 1 metal because: ◦ J forms a chloride containing the J^+ ion ◦ J forms a white chloride ◦ J has relatively low melting point, density and hardness ◦ J has a very fast reaction with water • some data does not fit: ◦ G forms a white compound ◦ electrical conductivity does not show the same pattern			
Total		6		

Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
09.1	to check they are safe (to consume)		1	AO2 3.4.1b	E
09.2	M contains A and C M does not contain B and D M contains an unknown compound	if no other mark awarded allow 1 mark for contains (at least) four compounds	1 1 1	AO3 3.4.1d	E
09.3	contains (only) one compound / element		1	AO1 3.4.1a	E
09.4	(distance moved by N ⇒) 2.0 (cm) and (distance moved by solvent ⇒) 4.4 cm $R_f \text{ value} = \frac{2.0}{4.4}$ = 0.45	an answer of 0.45 scores 3 marks allow value in the range 1.9 – 2.1 (cm) allow value in the range 4.3 – 4.5 (cm) allow correct calculation using student's measurements if within tolerance	1 1 1	AO2 3.4.1d	E

09.5	R _f value will be lower (because) spot will travel less far up paper		1 1	AO2 3.4.1e	E
09.6	R _f value will not change (because) the distances moved by N and the solvent increase by the same proportion		1 1	AO2 3.4.1e	E
Total			12		